

Dear Google Team,

I am applying for the Product Engineer, Global Manufacturing Engineering role. Google's work integrating advanced hardware across global data center infrastructure is a strong match for my experience taking complex electromechanical products from design through manufacturing, commercialization, and field support.

At Proteor, I lead product engineering for the ICARUS Lite, including custom dual voltage AC heaters that increased internal production by **30 percent**. I manage design documentation, BOM control, risk analysis, supplier coordination, and production support while collaborating across engineering, operations, sales, and customer service. I also developed a LabVIEW system that collects nine data points every 15 milliseconds from multiple laser sensors, enabling tolerance analysis, manufacturing yield monitoring, and quality control.

Previously, as CTO of Filament Innovations, I led the development and launch of the Gen3 POSEIDON, integrating industrial servo motion, dual extrusion, custom gantry hardware, electrical systems, and firmware. I also developed high speed linear motor systems that moved a 30 kilogram payload at up to 3 meters per second while maintaining 3 micron accuracy. This work required validating performance against targets, investigating root causes, coordinating overseas manufacturing partners, and driving corrective actions through deployment.

My progression from engineering technician to CTO and now Lead R&D Engineer reflects my ability to make rapid technical decisions as priorities change, collaborate directly with manufacturers, and deliver reliable products under demanding schedules.

Thank you for your consideration. I would welcome the opportunity to discuss how my product engineering and manufacturing experience can support Google's data center hardware programs.

Riley Beenders